

```
!pip install chromadb
!pip install sentence-transformers
!pip install langchain
```

```
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.17.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.13.0)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers>=4.34.0)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers>=4.34.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers>=4.34.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers>=4.34.0)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers>=2.2.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers>=2.2.2)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.24.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.24.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.24.0)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.24.0)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py)
Requirement already satisfied: langchain in /usr/local/lib/python3.12/dist-packages (1.2.14)
Requirement already satisfied: langchain-core<2.0.0,>=1.2.10 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: langgraph<1.2.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: langgraph-checkpoint<5.0.0,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: langgraph-sdk<0.4.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: xxhash>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from langchain)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch)
Requirement already satisfied: ormsgpack>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from langsmith)
Requirement already satisfied: httpx>=0.25.2 in /usr/local/lib/python3.12/dist-packages (from langsmith)
Requirement already satisfied: orjson>=3.11.5 in /usr/local/lib/python3.12/dist-packages (from langsmith)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith)
Requirement already satisfied: requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith)
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx)
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Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests)
```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or data
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub
modules.json: 100% 349/349 [00:00<00:00, 18.8kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 4.93kB/s]

README.md: 10.5k/? [00:00<00:00, 465kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.51kB/s]

config.json: 100% 612/612 [00:00<00:00, 43.5kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 457MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 467.14it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+---
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect
tokenizer_config.json: 100% 350/350 [00:00<00:00, 18.6kB/s]

vocab.txt: 232k/? [00:00<00:00, 6.52MB/s]

tokenizer.json: 466k/? [00:00<00:00, 21.8MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 6.95kB/s]

config.json: 100% 190/190 [00:00<00:00, 18.3kB/s]
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]

ids = ["doc1", "doc2", "doc3", "doc4", "doc5"]
```

```
embeddings = model.encode(documents)
```

```
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)
```

```
query = "What is AI?"

query_embedding = model.encode([query])
```

```
results = collection.query(
    query_embeddings=query_embedding,
    n_results=2
)

print(results)
```

```
{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence is the si
```

```
for doc in results['documents'][0]:
    print(doc)
```

```
Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.
```

```
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(query_embeddings=query_embedding, n_results=2)
    return results['documents'][0]

print(retrieve("Explain machine learning"))
```

```
['Machine Learning is a subset of AI.', 'Deep Learning uses neural networks.']
```

```
documents.extend([
    "Neural networks are inspired by the human brain.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds hidden patterns.",
    "Reinforcement learning learns via rewards."
])
```

Start coding or [generate](#) with AI.